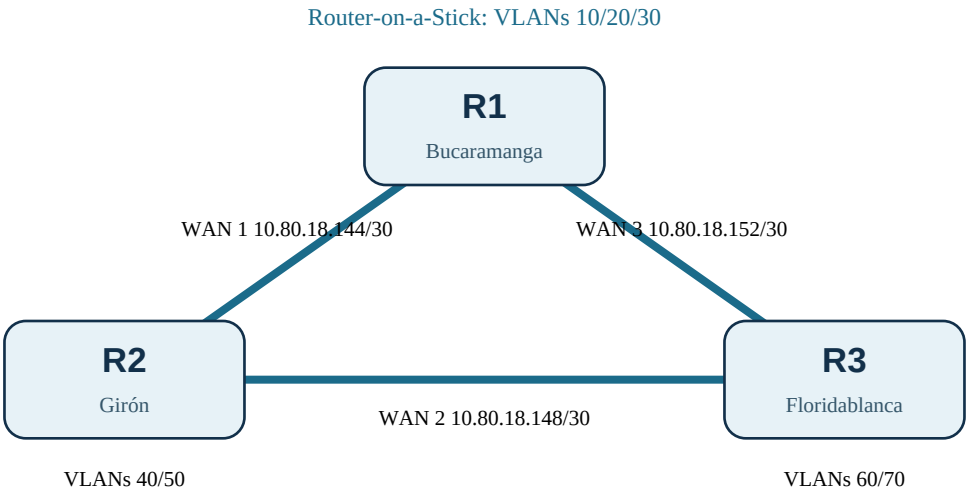
Lab 07 CCNA Enterprise VLSM

VLANs, enlaces troncales 802.1Q, Router-on-a-Stick y routing estatico triangular



Repositorio destino
CCNA-200-301-Networking-Labs / Lab 07
Estado
Configuracion verificada y documentada con evidencia de Packet Tracer

Resumen del laboratorio

Este laboratorio integra segmentacion por VLAN, VLSM, enlaces troncales 802.1Q, inter-VLAN routing mediante Router-on-a-Stick y tres enlaces WAN /30 entre Bucaramanga, Giron y Floridablanca. La topologia usa rutas estaticas directas entre sedes y exige rutas de ida y retorno para que ICMP funcione de extremo a extremo.

Resultado principal

Las tablas de routing revisadas contienen las rutas estaticas hacia las LAN remotas. La evidencia disponible confirma conectividad intersede Bucaramanga-Giron y el proceso de configuracion de Giron-Floridablanca. Se separan las pruebas exitosas de los primeros intentos con perdida inicial de paquetes.

Objetivos

Objetivo	Criterio de verificacion
VLSM	Siete LAN con prefijos dimensionados por necesidad
VLANs	VLAN 10/20/30/40/50/60/70 creadas y asignadas
Trunks	802.1Q entre switches y routers
WAN	Tres enlaces punto a punto /30
Routing	Rutas estaticas con next-hop vecino y retorno
Pruebas	ICMP local e intersede documentado

Plan de direccionamiento VLSM

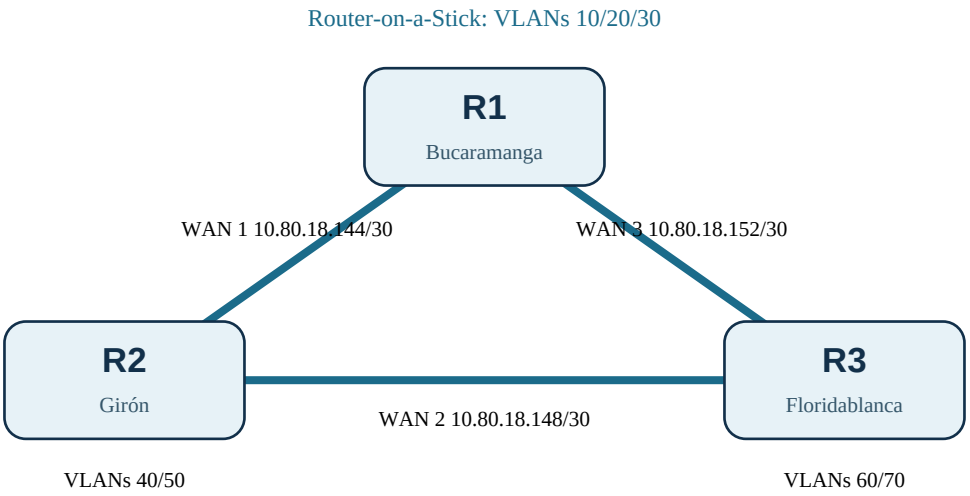
Sede	VLAN / Area	Red	Gateway
Bucaramanga	10 Desarrollo	10.80.16.0/24	10.80.16.1
Bucaramanga	20 Ingenieria	10.80.17.0/25	10.80.17.1
Bucaramanga	30 Administracion	10.80.18.0/26	10.80.18.1
Giron	40 RR. HH.	10.80.18.96/27	10.80.18.97
Giron	50 Gerencia	10.80.18.128/28	10.80.18.129
Floridablanca	60 Soporte TI	10.80.17.128/25	10.80.17.129
Floridablanca	70 Finanzas	10.80.18.64/27	10.80.18.65

Enlaces WAN

Enlace	Red /30	Extremo A	Extremo B
WAN 1 Bucaramanga-Giron	10.80.18.144/30	R1 .145	R2 .146
WAN 2 Giron-Floridablanca	10.80.18.148/30	R2 .149	R3 .150
WAN 3 Bucaramanga-Floridablanca	10.80.18.152/30	R1 .153	R3 .154

Correccion tecnica del diagrama WAN 3: el enlace correcto es 10.80.18.152/30, con R1 = 10.80.18.153 y R3 = 10.80.18.154. No debe confundirse con WAN 2, que usa 10.80.18.148/30.

Topologia y configuraciones verificadas



La arquitectura es triangular: Bucaramanga-Giron por WAN 1, Giron-Floridablanca por WAN 2 y Bucaramanga-Floridablanca por WAN 3. Las LAN se entregan mediante subinterfaces del router y encapsulacion dot1Q.

Comandos representativos

Componente	Configuracion verificada
VLANs	vlan 10, 20, 30, 40, 50, 60, 70
Trunk	switchport mode trunk; switchport trunk allowed vlan 10,20,30,40,50,60,70
Router-on-a-Stick	interface G0/x.; encapsulation dot1Q ; ip address
Rutas estaticas R2	10.80.16.0/24, 10.80.17.0/25, 10.80.18.0/26 via 10.80.18.145; redes de R3 via 10.80.18.150
Rutas estaticas R3	redes de R1 via 10.80.18.153; redes de R2 via 10.80.18.149
Persistencia	copy running-config startup-config

Rutas estaticas y razonamiento

R2 debe usar 10.80.18.150 como next hop para llegar a Soporte TI 10.80.17.128/25 y Finanzas 10.80.18.64/27. R3 debe usar 10.80.18.149 para regresar a RR. HH. 10.80.18.96/27 y Gerencia 10.80.18.128/28. Las rutas de retorno son necesarias porque el Echo Reply sigue el camino inverso.

Router	Destino	Next hop	Interpretacion
R2	10.80.16.0/24	10.80.18.145	LAN Desarrollo detras de R1
R2	10.80.17.0/25	10.80.18.145	LAN Ingenieria detras de R1
R2	10.80.18.0/26	10.80.18.145	LAN Administracion detras de R1
R2	10.80.17.128/25	10.80.18.150	LAN Soporte TI detras de R3
R2	10.80.18.64/27	10.80.18.150	LAN Finanzas detras de R3
R3	10.80.16.0/24	10.80.18.153	LAN Desarrollo detras de R1
R3	10.80.18.96/27	10.80.18.149	LAN RR. HH. detras de R2

Nota: una ruta marcada con S demuestra que esta instalada en la tabla; la prueba ICMP confirma adicionalmente que el reenvio y el retorno funcionan de extremo a extremo.

Matriz de pruebas y troubleshooting

Prueba	Resultado documentado	Clasificacion
PC9 Desarrollo -> PC0 RR. HH.	3/4 respuestas, TTL 126; perdida inicial 25%	Exito con perdida inicial de ARP
R1/R2 hacia redes remotas	Rutas S visibles en show ip route	Exito de control
R2 <-> R3	WAN 2 10.80.18.148/30 y next hops .149/.150	Configuracion verificada
PC0 Giron -> PC15 Soporte TI	Debe repetirse tras completar rutas de retorno	Pendiente de evidencia final
PC15 Soporte TI -> PC9 Desarrollo	Valida retorno desde Floridablanca	Pendiente de evidencia final

Interpretacion de pings fallidos

El resultado 3/4 no se presenta como fallo persistente: corresponde a una perdida inicial compatible con resolucion ARP durante el primer intento. Para declarar una prueba definitivamente fallida se debe repetir el ping y observar perdida continua. En este informe, las pruebas con evidencia positiva se etiquetan como exitosas y las no evidenciadas se dejan como pendientes, sin inventar resultados.

Troubleshooting aplicado

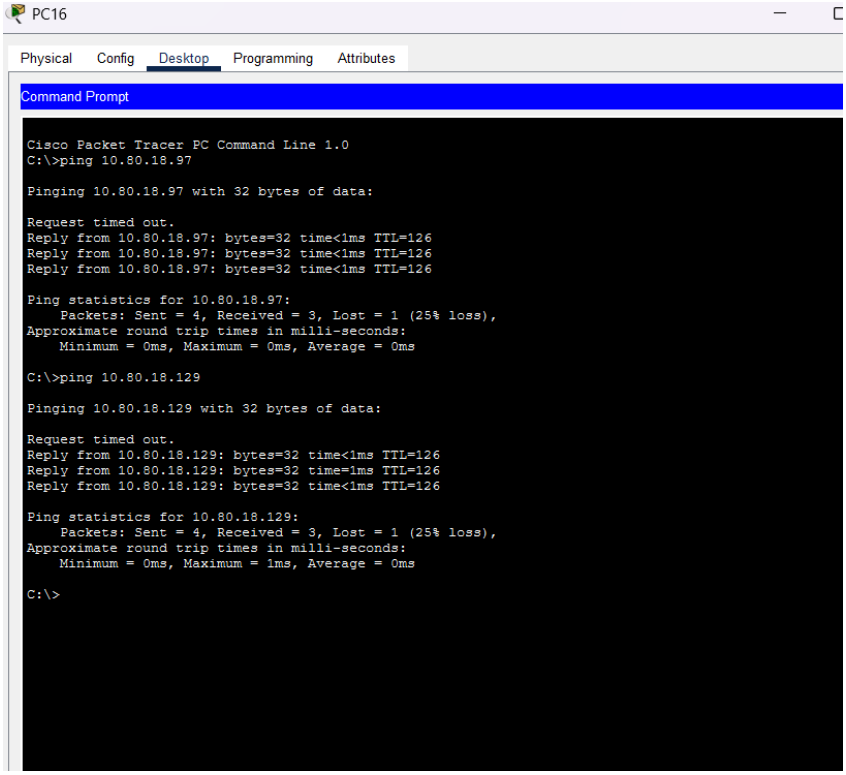
Sintoma	Causa probable	Accion
Rutas instaladas pero sin ping	Falta ruta de retorno	Agregar ruta estatica en el router remoto
Primer ping perdido	ARP inicial	Repetir ping y comparar resultados
WAN 3 inconsistente	Direccionamiento /30 mal rotulado	Usar 10.80.18.152/30: .153 y .154
VLAN local sin conectividad	Trunk o subinterfaz	Verificar show interfaces trunk y encapsulation dot1Q

Evidencia documental y configuraciones

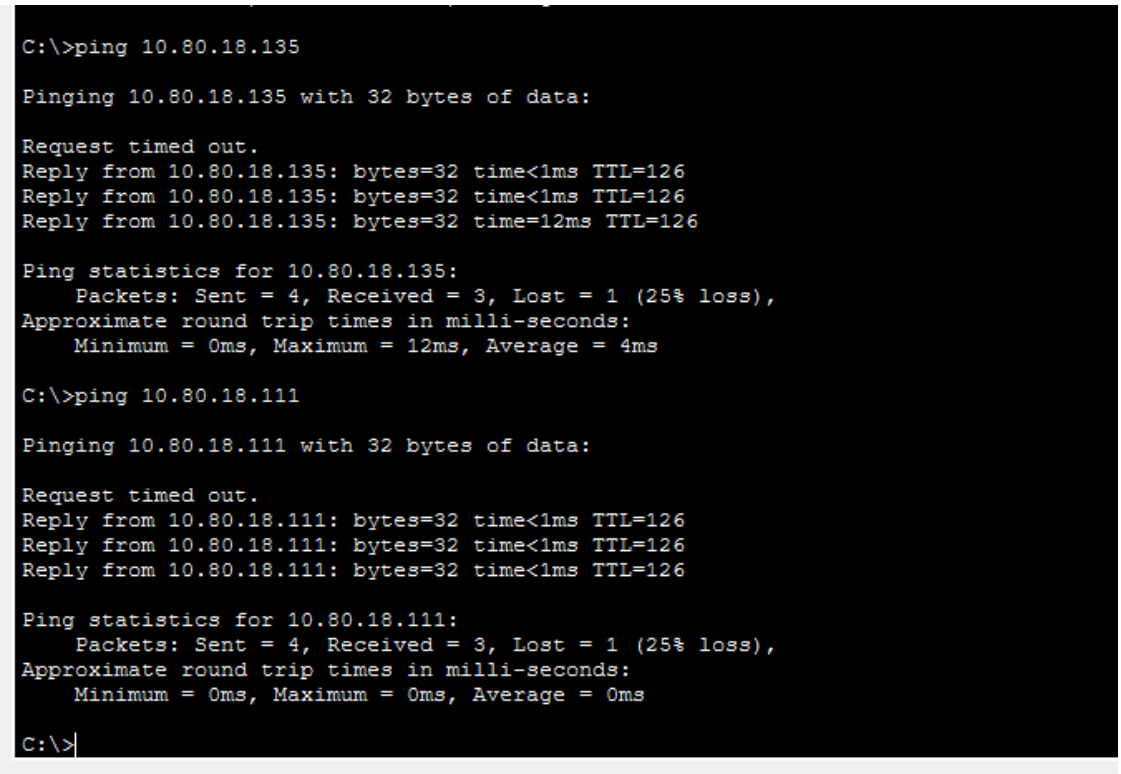
Las capturas 01 a 10 corresponden a las pruebas y configuraciones recuperadas del cierre del laboratorio. Las capturas 11 a 22 son la nueva evidencia aportada: incluyen el planteamiento VLSM en Excalidraw, VLANs activas, trunks 802.1Q, interfaces y tablas de routing de R1, R2, R3, SW1, SW2 y SW3.

Lectura tecnica del planteamiento

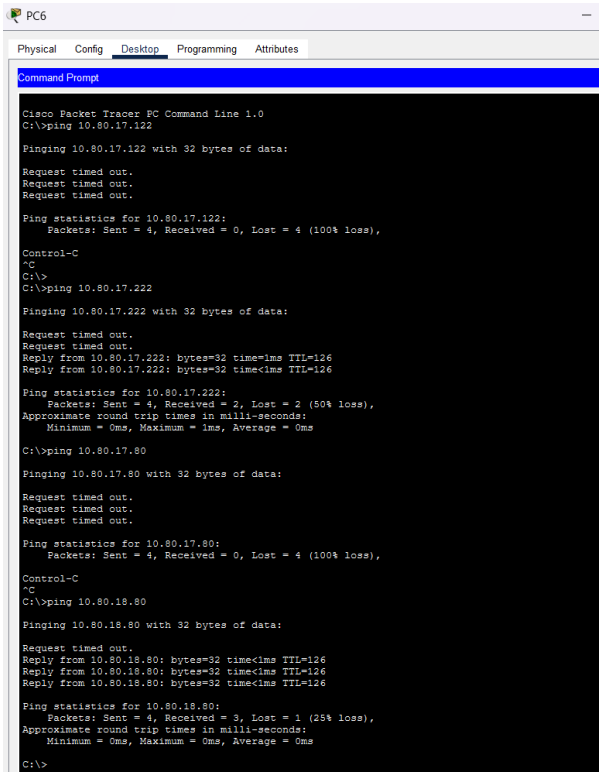
El planteamiento confirma el bloque 10.80.16.0/22, las siete LAN y los tres enlaces punto a punto. En la tarjeta de WAN 3 del dibujo hay un error de calculo: una red /30 tiene red .152, hosts .153 y .154, y broadcast .155. La documentacion final usa esa correccion y no los valores inconsistentes de la anotacion original.



Captura 01 - Evidencia del Lab 07 tomada durante la configuracion o validacion.



Captura 02 - Evidencia del Lab 07 tomada durante la configuracion o validacion.



Captura 03 - Evidencia del Lab 07 tomada durante la configuracion o validacion.

```
R3#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R3(config)#
R3(config)#ip route 10.80.18.96 255.255.255.224 10.80.18.149
R3(config)#ip route 10.80.18.128 255.255.255.240 10.80.18.149
R3(config)#
R3(config)#end
R3#show ip route
%SYS-5-CONFIG_I: Configured from console by console

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/8 is variably subnetted, 13 subnets, 7 masks
S       10.80.16.0/24 [1/0] via 10.80.18.153
S       10.80.17.0/25 [1/0] via 10.80.18.153
C       10.80.17.128/25 is directly connected, GigabitEthernet0/2.60
L       10.80.17.254/32 is directly connected, GigabitEthernet0/2.60
S       10.80.18.0/26 [1/0] via 10.80.18.153
C       10.80.18.64/27 is directly connected, GigabitEthernet0/2.70
L       10.80.18.94/32 is directly connected, GigabitEthernet0/2.70
S       10.80.18.96/27 [1/0] via 10.80.18.149
S       10.80.18.128/28 [1/0] via 10.80.18.149
C       10.80.18.148/30 is directly connected, GigabitEthernet0/1
L       10.80.18.150/32 is directly connected, GigabitEthernet0/1
C       10.80.18.152/30 is directly connected, GigabitEthernet0/0
L       10.80.18.154/32 is directly connected, GigabitEthernet0/0

R3#
R3#
```

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Captura 04 - Evidencia del Lab 07 tomada durante la configuracion o validacion.

```
Building configuration...
[OK]
R2#
R2#enable
R2#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R2(config)#
R2(config)#ip route 10.80.17.128 255.255.255.128 10.80.18.150
R2(config)#ip route 10.80.18.64 255.255.255.224 10.80.18.150
R2(config)#
R2(config)#end
R2#show ip route
%SYS-5-CONFIG_I: Configured from console by console

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/8 is variably subnetted, 13 subnets, 7 masks
S       10.80.16.0/24 [1/0] via 10.80.18.145
S       10.80.17.0/25 [1/0] via 10.80.18.145
S       10.80.17.128/25 [1/0] via 10.80.18.150
S       10.80.18.0/26 [1/0] via 10.80.18.145
S       10.80.18.64/27 [1/0] via 10.80.18.150
C       10.80.18.96/27 is directly connected, GigabitEthernet0/0.40
L       10.80.18.126/32 is directly connected, GigabitEthernet0/0.40
C       10.80.18.128/28 is directly connected, GigabitEthernet0/0.50
L       10.80.18.142/32 is directly connected, GigabitEthernet0/0.50
C       10.80.18.144/30 is directly connected, GigabitEthernet0/1
L       10.80.18.146/32 is directly connected, GigabitEthernet0/1
C       10.80.18.148/30 is directly connected, GigabitEthernet0/2
L       10.80.18.149/32 is directly connected, GigabitEthernet0/2

R2#
R2#
```

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Captura 05 - Evidencia del Lab 07 tomada durante la configuracion o validacion.

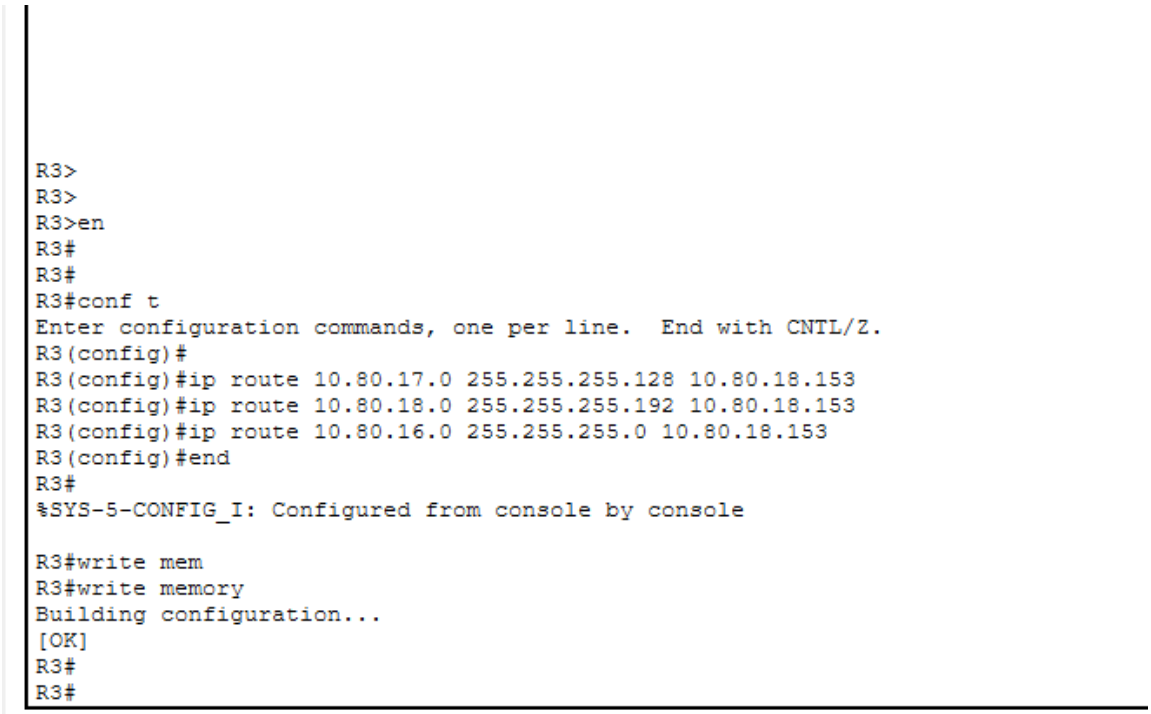
```
R3#sh ip rou
R3#sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/8 is variably subnetted, 11 subnets, 6 masks
S       10.80.16.0/24 [1/0] via 10.80.18.153
S       10.80.17.0/25 [1/0] via 10.80.18.153
C       10.80.17.128/25 is directly connected, GigabitEthernet0/2.60
L       10.80.17.254/32 is directly connected, GigabitEthernet0/2.60
S       10.80.18.0/26 [1/0] via 10.80.18.153
C       10.80.18.64/27 is directly connected, GigabitEthernet0/2.70
L       10.80.18.94/32 is directly connected, GigabitEthernet0/2.70
C       10.80.18.148/30 is directly connected, GigabitEthernet0/1
L       10.80.18.150/32 is directly connected, GigabitEthernet0/1
C       10.80.18.152/30 is directly connected, GigabitEthernet0/0
L       10.80.18.154/32 is directly connected, GigabitEthernet0/0

R3#
```

Captura 06 - Evidencia del Lab 07 tomada durante la configuracion o validacion.



```
R3>
R3>
R3>en
R3#
R3#
R3#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
R3(config)#
R3(config)#ip route 10.80.17.0 255.255.255.128 10.80.18.153
R3(config)#ip route 10.80.18.0 255.255.255.192 10.80.18.153
R3(config)#ip route 10.80.16.0 255.255.255.0 10.80.18.153
R3(config)#end
R3#
%SYS-5-CONFIG_I: Configured from console by console

R3#write mem
R3#write memory
Building configuration...
[OK]
R3#
R3#
```

Captura 07 - Evidencia del Lab 07 tomada durante la configuracion o validacion.

```
L       10.80.18.153/32 is directly connected, GigabitEthernet0/1

R1(config)#end
R1#
%SYS-5-CONFIG_I: Configured from console by console

R1#
R1#wr
R1#write memo
R1#write memory
Building configuration...
[OK]
R1#
R1#
R1#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#ip route 10.80.17.128 255.255.255.128 10.80.18.154
R1(config)#ip route 10.80.18.64 255.255.255.224 10.80.18.154
R1(config)#end
R1#show ip route
%SYS-5-CONFIG_I: Configured from console by console

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/8 is variably subnetted, 14 subnets, 7 masks
C       10.80.16.0/24 is directly connected, GigabitEthernet0/2.30
L       10.80.16.254/32 is directly connected, GigabitEthernet0/2.30
C       10.80.17.0/25 is directly connected, GigabitEthernet0/2.10
L       10.80.17.126/32 is directly connected, GigabitEthernet0/2.10
S       10.80.17.128/25 [1/0] via 10.80.18.154
C       10.80.18.0/26 is directly connected, GigabitEthernet0/2.20
L       10.80.18.62/32 is directly connected, GigabitEthernet0/2.20
S       10.80.18.64/27 [1/0] via 10.80.18.154
S       10.80.18.96/27 [1/0] via 10.80.18.146
S       10.80.18.128/28 [1/0] via 10.80.18.146
C       10.80.18.144/30 is directly connected, GigabitEthernet0/0
L       10.80.18.145/32 is directly connected, GigabitEthernet0/0
C       10.80.18.152/30 is directly connected, GigabitEthernet0/1
L       10.80.18.153/32 is directly connected, GigabitEthernet0/1

R1#write memo
R1#write memory
Building configuration...
[OK]
R1#
R1#
```

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Captura 08 - Evidencia del Lab 07 tomada durante la configuracion o validacion.

```
R1#write memo
R1#write memory
Building configuration...
[OK]
R1#
R1#
R1#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#ip route 10.80.17.128 255.255.255.128 10.80.18.154
R1(config)#ip route 10.80.18.64 255.255.255.224 10.80.18.154
R1(config)#end
R1#show ip route
%SYS-5-CONFIG_I: Configured from console by console

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

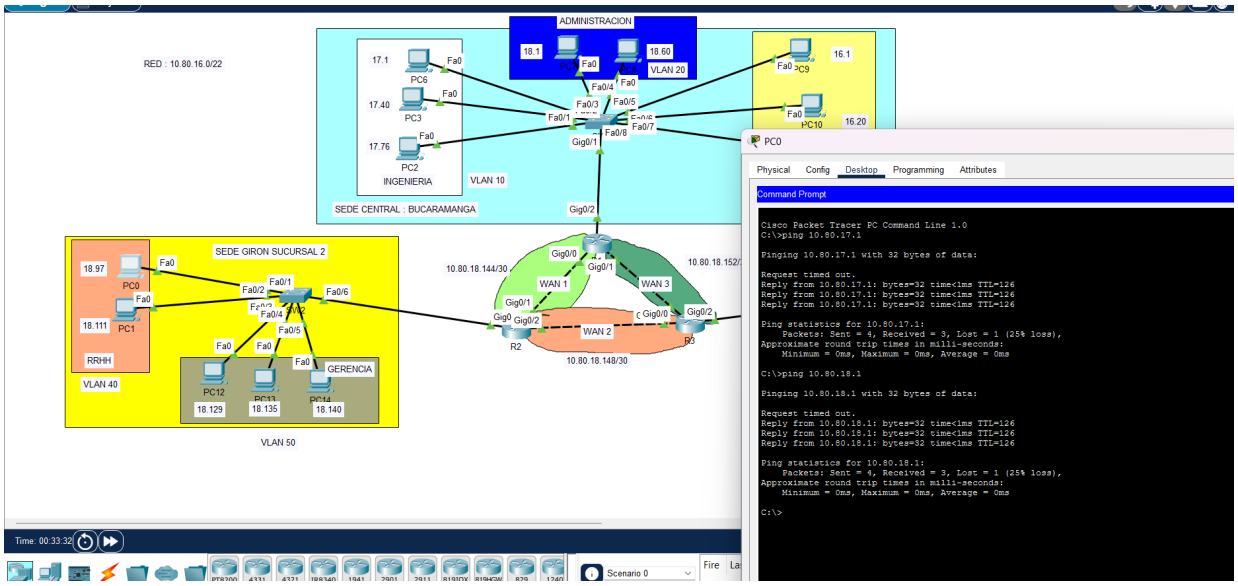
Gateway of last resort is not set

     10.0.0.0/8 is variably subnetted, 14 subnets, 7 masks
C       10.80.16.0/24 is directly connected, GigabitEthernet0/2.30
L       10.80.16.254/32 is directly connected, GigabitEthernet0/2.30
C       10.80.17.0/25 is directly connected, GigabitEthernet0/2.10
L       10.80.17.126/32 is directly connected, GigabitEthernet0/2.10
S       10.80.17.128/25 [1/0] via 10.80.18.154
C       10.80.18.0/26 is directly connected, GigabitEthernet0/2.20
L       10.80.18.62/32 is directly connected, GigabitEthernet0/2.20
S       10.80.18.64/27 [1/0] via 10.80.18.154
S       10.80.18.96/27 [1/0] via 10.80.18.146
S       10.80.18.128/28 [1/0] via 10.80.18.146
C       10.80.18.144/30 is directly connected, GigabitEthernet0/0
L       10.80.18.145/32 is directly connected, GigabitEthernet0/0
C       10.80.18.152/30 is directly connected, GigabitEthernet0/1
L       10.80.18.153/32 is directly connected, GigabitEthernet0/1

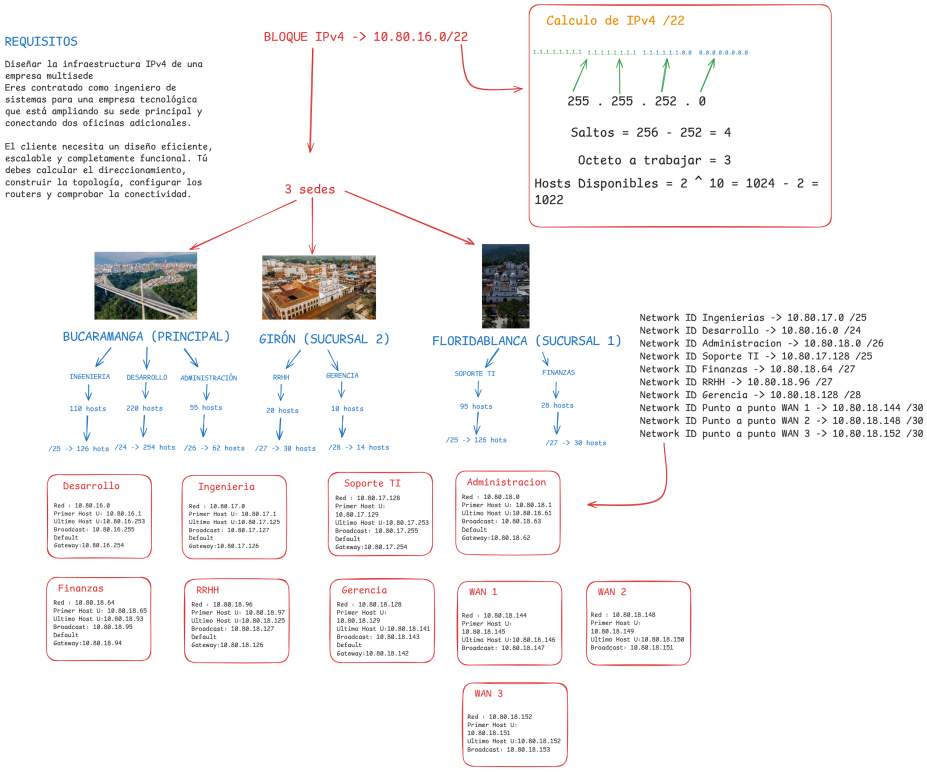
R1#
```

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Captura 09 - Evidencia del Lab 07 tomada durante la configuracion o validacion.



Captura 10 - Evidencia del Lab 07 tomada durante la configuracion o validacion.



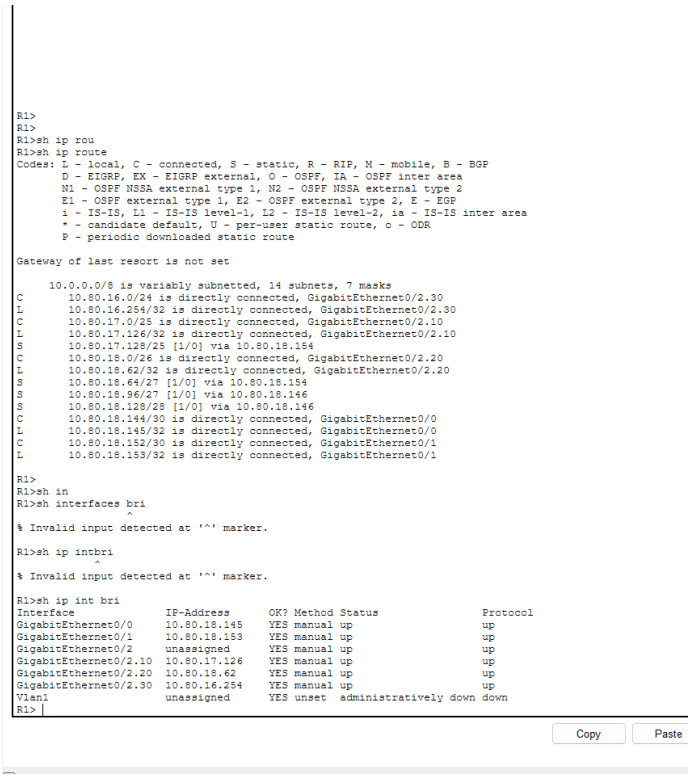
Captura 11 - Planteamiento VLSM y direccionamiento en Excalidraw.

```
SW1>
SW1>
SW1>sh vl
SW1>sh vlan bri
SW1>sh vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/9, Fa0/10, Fa0/11, Fa0/12
                                                Fa0/13, Fa0/14, Fa0/15, Fa0/16
                                                Fa0/17, Fa0/18, Fa0/19, Fa0/20
                                                Fa0/21, Fa0/22, Fa0/23, Fa0/24
                                                Gig0/2
10   ENGINEERING                      active    Fa0/1, Fa0/2, Fa0/3
20   ADMINISTRATION                   active    Fa0/4, Fa0/5
30   DEVELOPMENT                      active    Fa0/6, Fa0/7, Fa0/8
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
SW1>
SW1>
```

Copy P

Captura 12 - Configuracion de switches, routers y routing estatico.



```
R1>
R1>
R1>sh ip rou
R1>sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/8 is variably subnetted, 14 subnets, 7 masks
C       10.80.16.0/24 is directly connected, GigabitEthernet0/2.30
L       10.80.16.254/32 is directly connected, GigabitEthernet0/2.30
C       10.80.17.0/25 is directly connected, GigabitEthernet0/2.10
L       10.80.17.126/32 is directly connected, GigabitEthernet0/2.10
S       10.80.17.128/25 [1/0] via 10.80.18.154
C       10.80.18.0/26 is directly connected, GigabitEthernet0/2.20
L       10.80.18.62/32 is directly connected, GigabitEthernet0/2.20
S       10.80.18.64/27 [1/0] via 10.80.18.154
S       10.80.18.96/27 [1/0] via 10.80.18.146
S       10.80.18.128/28 [1/0] via 10.80.18.146
C       10.80.18.144/30 is directly connected, GigabitEthernet0/0
L       10.80.18.145/32 is directly connected, GigabitEthernet0/0
C       10.80.18.152/30 is directly connected, GigabitEthernet0/1
L       10.80.18.153/32 is directly connected, GigabitEthernet0/1

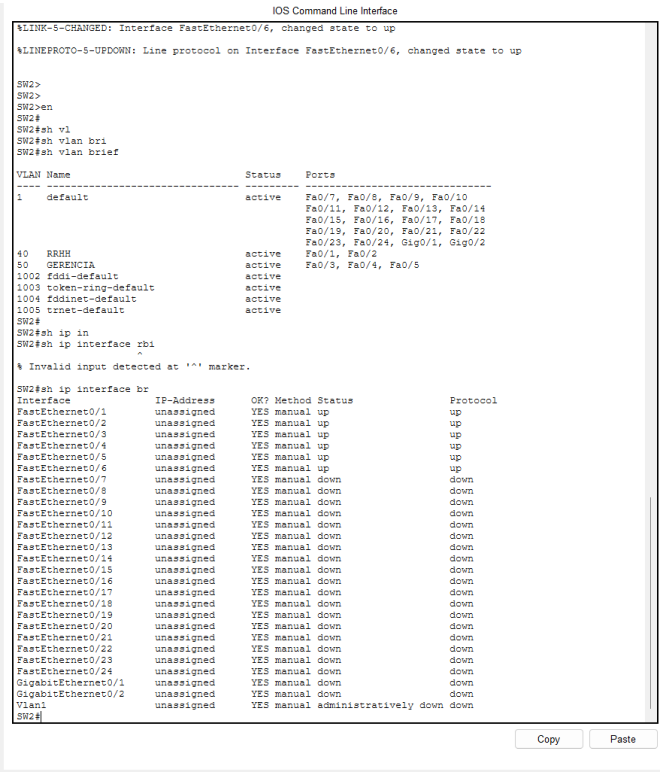
R1>
R1>sh in
R1>sh interfaces bri
                 ^
% Invalid input detected at '^' marker.

R1>sh ip intbri
              ^
% Invalid input detected at '^' marker.

R1>sh ip int bri
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     10.80.18.145    YES manual up                    up
GigabitEthernet0/1     10.80.18.153    YES manual up                    up
GigabitEthernet0/2     unassigned      YES manual up                    up
GigabitEthernet0/2.10  10.80.17.126    YES manual up                    up
GigabitEthernet0/2.20  10.80.18.62     YES manual up                    up
GigabitEthernet0/2.30  10.80.16.254    YES manual up                    up
Vlan1                  unassigned      YES unset  administratively down down
R1>
```

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Captura 13 - Configuracion de switches, routers y routing estatico.

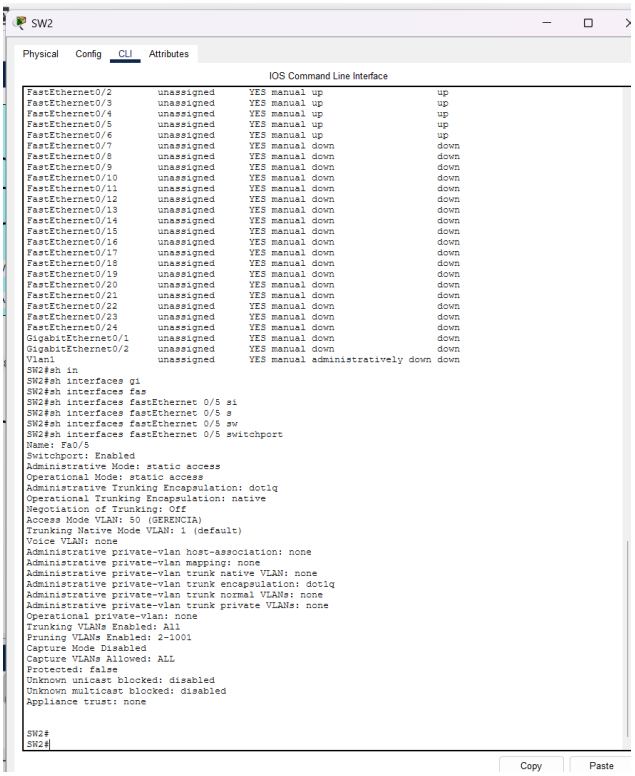


Captura 14 - Configuracion de switches, routers y routing estatico.

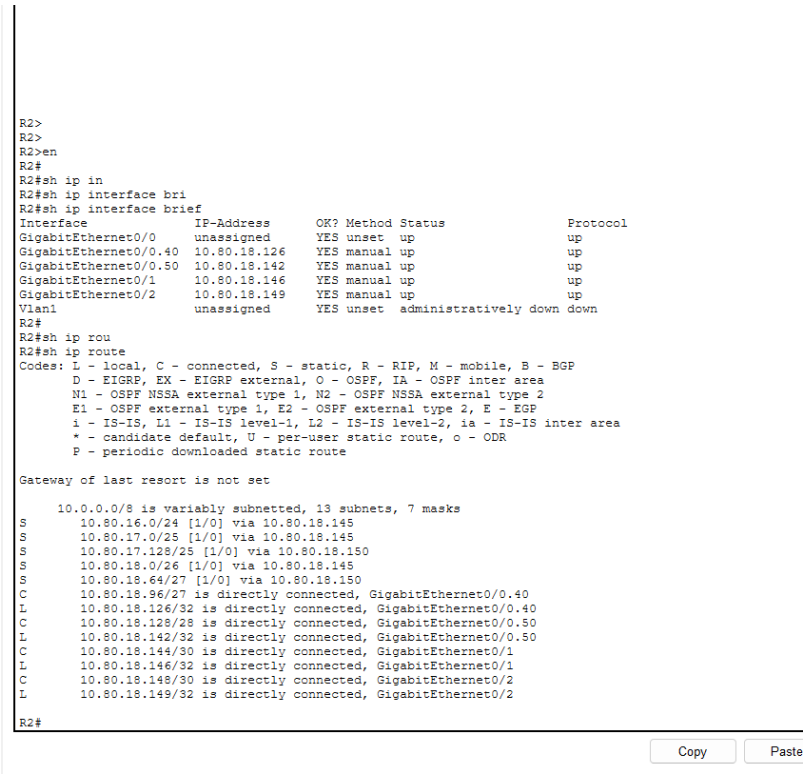
```
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
SW1>
SW1>
SW1>
SW1>en
SW1#
SW1#sh in
SW1#sh interfaces gi
SW1#sh interfaces gigabitEthernet g
SW1#sh interfaces gigabitEthernet 0/1 sw
SW1#sh interfaces gigabitEthernet 0/1 switchport
Name: Gig0/1
Switchport: Enabled
Administrative Mode: trunk
Operational Mode: trunk
Administrative Trunking Encapsulation: dot1q
Operational Trunking Encapsulation: dot1q
Negotiation of Trunking: On
Access Mode VLAN: 1 (default)
Trunking Native Mode VLAN: 1 (default)
Voice VLAN: none
Administrative private-vlan host-association: none
Administrative private-vlan mapping: none
Administrative private-vlan trunk native VLAN: none
Administrative private-vlan trunk encapsulation: dot1q
Administrative private-vlan trunk normal VLANs: none
Administrative private-vlan trunk private VLANs: none
Operational private-vlan: none
Trunking VLANs Enabled: 10,20,30
Pruning VLANs Enabled: 2-1001
Capture Mode Disabled
Capture VLANs Allowed: ALL
Protected: false
Unknown unicast blocked: disabled
Unknown multicast blocked: disabled
Appliance trust: none


SW1#
```

Captura 15 - Configuracion de switches, routers y routing estatico.



Captura 16 - Configuracion de switches, routers y routing estatico.



```
R2>
R2>
R2>en
R2#
R2#sh ip in
R2#sh ip interface bri
R2#sh ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     unassigned      YES unset  up                    up
GigabitEthernet0/0.40  10.80.18.126    YES manual up                    up
GigabitEthernet0/0.50  10.80.18.142    YES manual up                    up
GigabitEthernet0/1     10.80.18.146    YES manual up                    up
GigabitEthernet0/2     10.80.18.149    YES manual up                    up
Vlan1                  unassigned      YES unset  administratively down down
R2#
R2#sh ip rou
R2#sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/8 is variably subnetted, 13 subnets, 7 masks
S       10.80.16.0/24 [1/0] via 10.80.18.145
S       10.80.17.0/25 [1/0] via 10.80.18.145
S       10.80.17.128/25 [1/0] via 10.80.18.150
S       10.80.18.0/26 [1/0] via 10.80.18.145
S       10.80.18.64/27 [1/0] via 10.80.18.150
C       10.80.18.96/27 is directly connected, GigabitEthernet0/0.40
L       10.80.18.126/32 is directly connected, GigabitEthernet0/0.40
C       10.80.18.128/28 is directly connected, GigabitEthernet0/0.50
L       10.80.18.142/32 is directly connected, GigabitEthernet0/0.50
C       10.80.18.144/30 is directly connected, GigabitEthernet0/1
L       10.80.18.146/32 is directly connected, GigabitEthernet0/1
C       10.80.18.148/30 is directly connected, GigabitEthernet0/2
L       10.80.18.149/32 is directly connected, GigabitEthernet0/2

R2#
```

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Captura 17 - Configuracion de switches, routers y routing estatico.

```
R2>
R2>
R2>en
R2#
R2#sh ip in
R2#sh ip interface bri
R2#sh ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     unassigned      YES unset  up                    up
GigabitEthernet0/0.40  10.80.18.126    YES manual up                    up
GigabitEthernet0/0.50  10.80.18.142    YES manual up                    up
GigabitEthernet0/1     10.80.18.146    YES manual up                    up
GigabitEthernet0/2     10.80.18.149    YES manual up                    up
Vlan1                  unassigned      YES unset  administratively down down
R2#
R2#sh ip rou
R2#sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

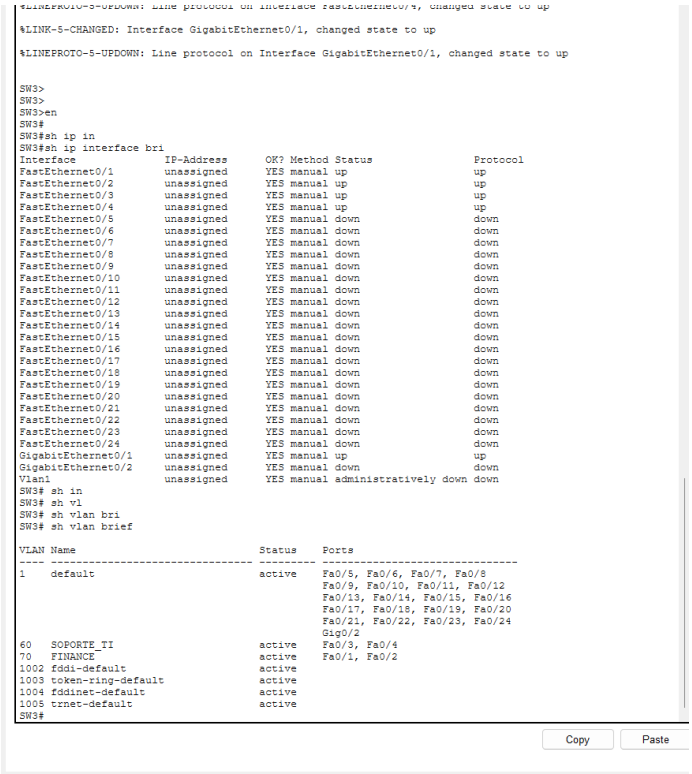
Gateway of last resort is not set

     10.0.0.0/8 is variably subnetted, 13 subnets, 7 masks
S       10.80.16.0/24 [1/0] via 10.80.18.145
S       10.80.17.0/25 [1/0] via 10.80.18.145
S       10.80.17.128/25 [1/0] via 10.80.18.150
S       10.80.18.0/26 [1/0] via 10.80.18.145
S       10.80.18.64/27 [1/0] via 10.80.18.150
C       10.80.18.96/27 is directly connected, GigabitEthernet0/0.40
L       10.80.18.126/32 is directly connected, GigabitEthernet0/0.40
C       10.80.18.128/28 is directly connected, GigabitEthernet0/0.50
L       10.80.18.142/32 is directly connected, GigabitEthernet0/0.50
C       10.80.18.144/30 is directly connected, GigabitEthernet0/1
L       10.80.18.146/32 is directly connected, GigabitEthernet0/1
C       10.80.18.148/30 is directly connected, GigabitEthernet0/2
L       10.80.18.149/32 is directly connected, GigabitEthernet0/2

R2#
```

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Captura 18 - Configuracion de switches, routers y routing estatico.

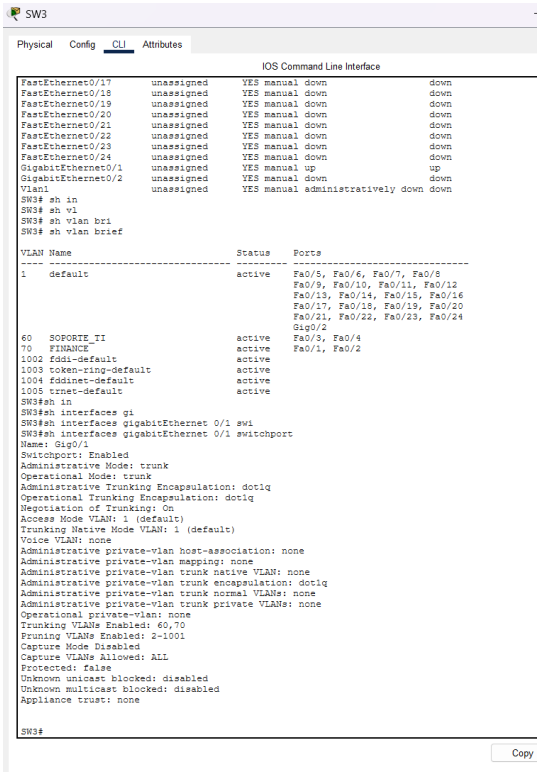


Captura 19 - Configuracion de switches, routers y routing estatico.

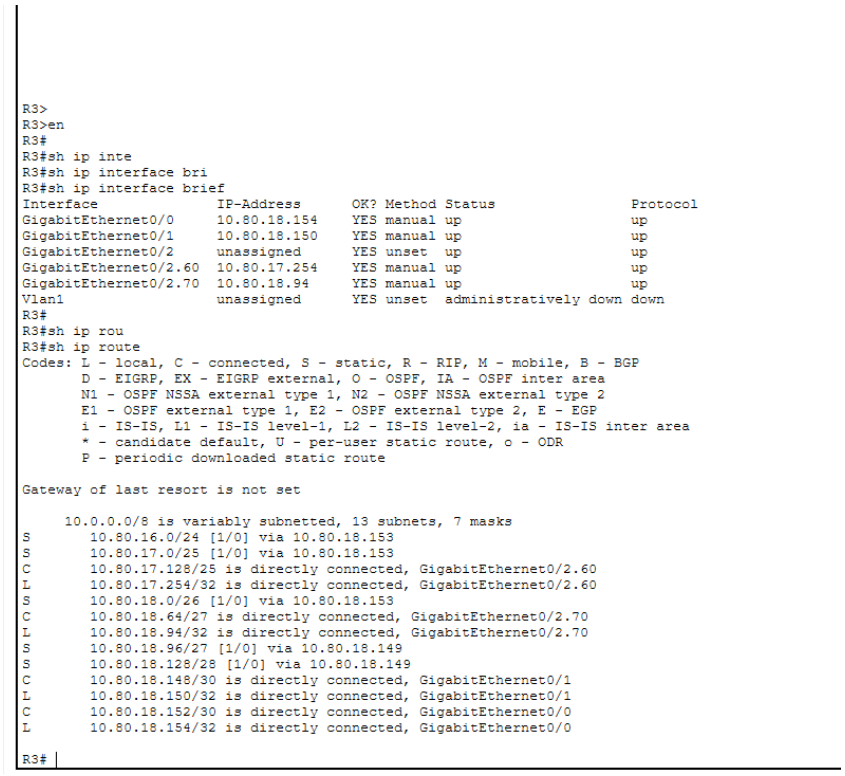
```
60   SOPORTE_TI                       active    Fa0/3, Fa0/4
70   FINANCE                          active    Fa0/1, Fa0/2
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
SW3#sh in
SW3#sh interfaces gi
SW3#sh interfaces gigabitEthernet 0/1 swi
SW3#sh interfaces gigabitEthernet 0/1 switchport
Name: Gig0/1
Switchport: Enabled
Administrative Mode: trunk
Operational Mode: trunk
Administrative Trunking Encapsulation: dot1q
Operational Trunking Encapsulation: dot1q
Negotiation of Trunking: On
Access Mode VLAN: 1 (default)
Trunking Native Mode VLAN: 1 (default)
Voice VLAN: none
Administrative private-vlan host-association: none
Administrative private-vlan mapping: none
Administrative private-vlan trunk native VLAN: none
Administrative private-vlan trunk encapsulation: dot1q
Administrative private-vlan trunk normal VLANs: none
Administrative private-vlan trunk private VLANs: none
Operational private-vlan: none
Trunking VLANs Enabled: 60,70
Pruning VLANs Enabled: 2-1001
Capture Mode Disabled
Capture VLANs Allowed: ALL
Protected: false
Unknown unicast blocked: disabled
Unknown multicast blocked: disabled
Appliance trust: none


SW3#
```

Captura 20 - Configuracion de switches, routers y routing estatico.



Captura 21 - Configuracion de switches, routers y routing estatico.



```
R3>
R3>en
R3#
R3#sh ip inte
R3#sh ip interface bri
R3#sh ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     10.80.18.154    YES manual up                    up
GigabitEthernet0/1     10.80.18.150    YES manual up                    up
GigabitEthernet0/2     unassigned      YES unset  up                    up
GigabitEthernet0/2.60  10.80.17.254    YES manual up                    up
GigabitEthernet0/2.70  10.80.18.94     YES manual up                    up
Vlan1                  unassigned      YES unset  administratively down down
R3#
R3#sh ip rou
R3#sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/8 is variably subnetted, 13 subnets, 7 masks
S       10.80.16.0/24 [1/0] via 10.80.18.153
S       10.80.17.0/25 [1/0] via 10.80.18.153
C       10.80.17.128/25 is directly connected, GigabitEthernet0/2.60
L       10.80.17.254/32 is directly connected, GigabitEthernet0/2.60
S       10.80.18.0/26 [1/0] via 10.80.18.153
C       10.80.18.64/27 is directly connected, GigabitEthernet0/2.70
L       10.80.18.94/32 is directly connected, GigabitEthernet0/2.70
S       10.80.18.96/27 [1/0] via 10.80.18.149
S       10.80.18.128/28 [1/0] via 10.80.18.149
C       10.80.18.148/30 is directly connected, GigabitEthernet0/1
L       10.80.18.150/32 is directly connected, GigabitEthernet0/1
C       10.80.18.152/30 is directly connected, GigabitEthernet0/0
L       10.80.18.154/32 is directly connected, GigabitEthernet0/0

R3#
```

Captura 22 - Configuracion de switches, routers y routing estatico.